EXP NO: 3

DATE: 23/7 LEARNING & ASSIGNMENT OF IP ADDRESS MANUALLY TO COMPUTERS.

AIM:

To understand the Ip addressing schemes and manually add Ip address to computer on network.

THEORY:

1) An Ip address is an unique numerical label assigned to each device connected to a computer network

2) IPV4 addresses are 32-bit numbers written in dotted-decimal notations

3) IPV6 uses 128-bit address

4) Subnet mask determines the network and host portions

5) Manual (Static) Ip attignment is used for severs and network devices requiring a permanent address

6) DHCP attigns Ip dynamically.

PROCEDURE:

1) In windows: open network and sharing center

2) Change Adapter settings → Right click Local Area Connection → properties → IPV4 → Set Ip manually

2) Enter Ip address: 192.168.1.1.0 Subnet Mask: 255.255.255.0, Default gateway: 192.168.1.1

3) Enter DNS server: 8.8.8.8 (Google DNS), click OK

4) In Linux: Use command 'Sudo ip addr add 192.168.1.10/24 dev etho'

5) To set gateway in Linux 'Sudo ip route add default via 192.168.1.1'

6) For permanent configuration in Linux, edit/ etc/ network/ interfaces (or) use nmtui.

7) verify the configuration using 'ipconfig' or 'ipa'

8) ping the default gateway to confirms network connectivity

9) Ping another computer on the same subnet to verify communication.

CODE:

```
sudo ip addr add 192.168.1.10/24 dev eth0
sudo ip route add default via 192.168.1.1
ip a show eth0
ping 192.168.1.1
nmcli con mod 'Wired connection 1' ipv4.addresses 192.168.1.10/24
nmcli con mod 'wired connection 1' ipv4.gateway 192.168.1.1
nmcli con mod 'wired connection 1' ipv4.method
nmcli con up 'wired connection 1' manual.
```

RESULT:

Static Ip address were successfully assigned to computers. Connectivity was verified using ping commands between hosts on the same subnet.